DUBAI MUNICIPALITY AI PARK DESIGN CHALLENGE

UPLOAD SLOT 05 OF 12

3D & Spatial Visualizations

Presentation boards and illustrative views. Renders are artistic impressions; the analysis outputs are computed.

CONTENTS

Al Safa 2 Park — Falaj Al Safa

Mohamed Wasim · Individual Applicant

Generated 2026-08-03 · reproducible: python tools/build_submission_pdfs.py

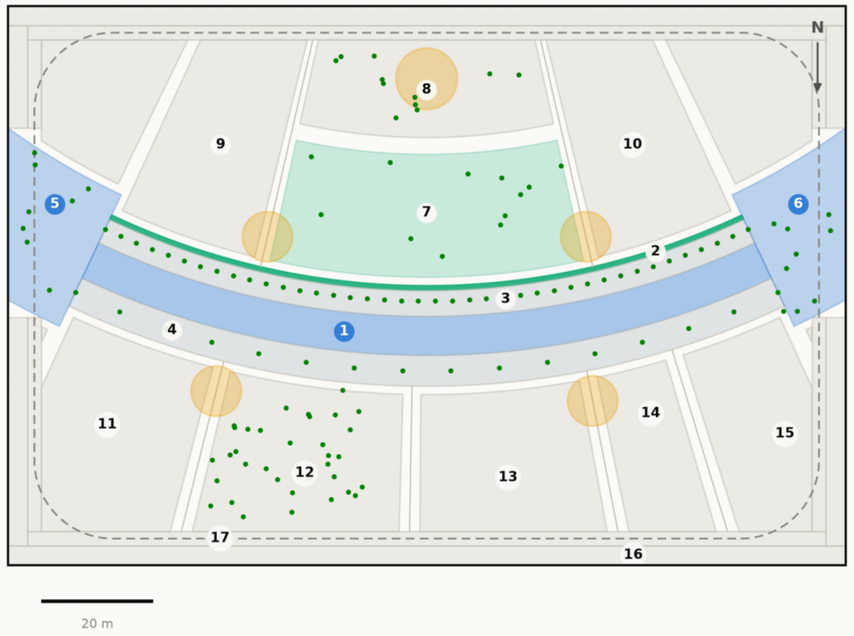
AL SAFA 2 PARK — FALAJ AL SAFA

Dubai Municipality AI Park Design Challenge | Board 1 of 2 — CONCEPT

The concept, the plan, and the section that makes it work

Masterplan — Falaj Al Safa

One arc, and every room struck off its centre. Drawn from src/plan.py, the same geometry the models use



ROOM SCHEDULE

1	Al Mamsha — the Crescent Walk	871 m²
2	Al Falaj — the water channel	105 m²
3	Crescent Shade Margin (N)	549 m²
4	Crescent Shade Margin (S)	715 m²
5	West Gate Majlis	439 m²
6	East Gate Majlis	439 m²
7	Al Nakhil — the Oasis Basin	1,140 m²
8	Quiet Contemplation Garden	707 m²
9	Children's Dune Play	1,267 m²
10	Family Picnic Grove	1,267 m²
11	Outdoor Fitness Terrace	908 m²
12	Native Planting / Biodiversity Wadi	894 m²
13	Community Plaza & Event Lawn	787 m²
14	Souk Kiosks & Services	417 m²
15	Multipurpose Sports Lawn	630 m²
16	Al Kathib — the dune berm	1,464 m²
17	Al Madar — the perimeter loop	986 m²
	Al Sikkak — shaded alleys & setbacks	1,414 m²
	Total	**15,000 m²**

Crescent radius 141 m, sagitta 18 m. Green dots are the 131 trees; circles are the majlis pavilions; the dashed line is Al Madar, the running loop. Site boundary is an ASSUMED 150×100 m rectangle pending DWG confirmation.
Source: src/plan.py — areas are the shoelace area of each drawn polygon · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

The crescent from the north-west, at golden hour

Visualisation in preparation

Aerial view · RENDER_PROMPTS.md prompt 01

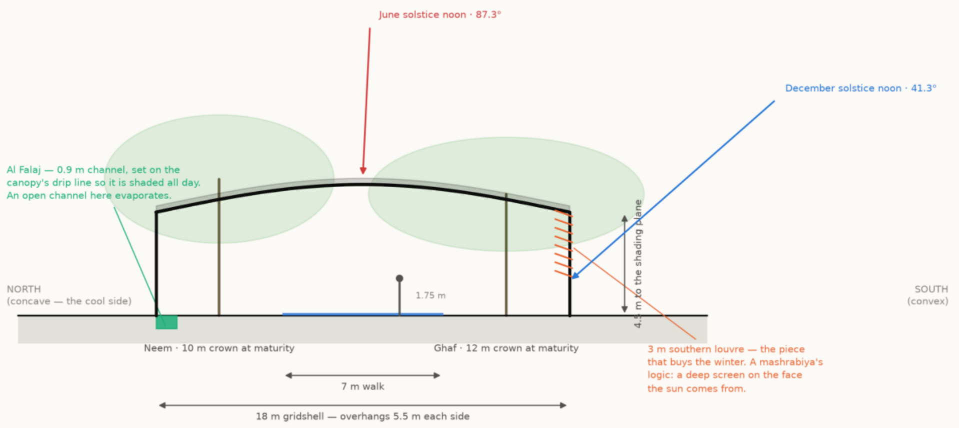
Beneath Al Hilal — the perforated soffit at eye level

Visualisation in preparation

Eye-level view · RENDER_PROMPTS.md prompt 03

Section A-A — the Crescent Canopy at midspan

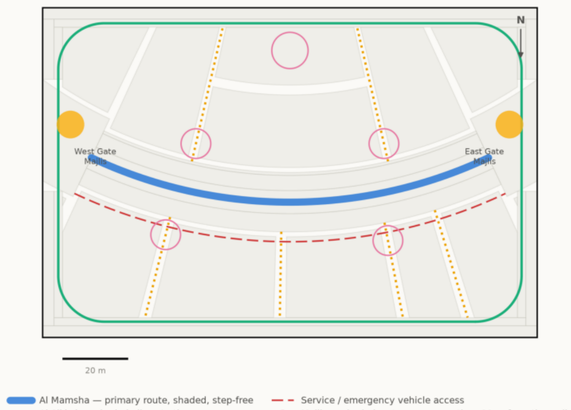
Every dimension read from config.CRESCENT; sun angles computed by the NREL algorithm at 25.190°N



The plane alone loses the walk to a low southern sun. The louvre is what holds the shadow on the path from November to January.
Source: src/config.py CRESCENT; sun angles NREL SPA via pvlib · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Circulation & accessibility

One shaded primary route, radial alleys off it, and a running loop around the whole



Every room is reached from the shaded route by one alley. The site is level, so the whole park is step-free; the only change of level in the scheme is the Oasis Basin, which is ramped.
Source: src/plan.py — routes are the drawn geometry, not a sketch over it · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

One arc. Every room in the park is struck off its centre, so no room is a rectangle and every room faces the crescent square-on.
The bow is 18 m deep on a 141 m radius — swept against the 8,760-hour solar model, not drawn by eye. A straight canopy shades marginally more ground on average and leaves the walk with no shade at all for 330 hours a year; this one leaves it for 56.
Walk shaded 87.3% of daylight hours · 131 trees · 15,000 m² · every number reproducible with `python run_analysis.py`

Board 2 evidence

Presentation board.

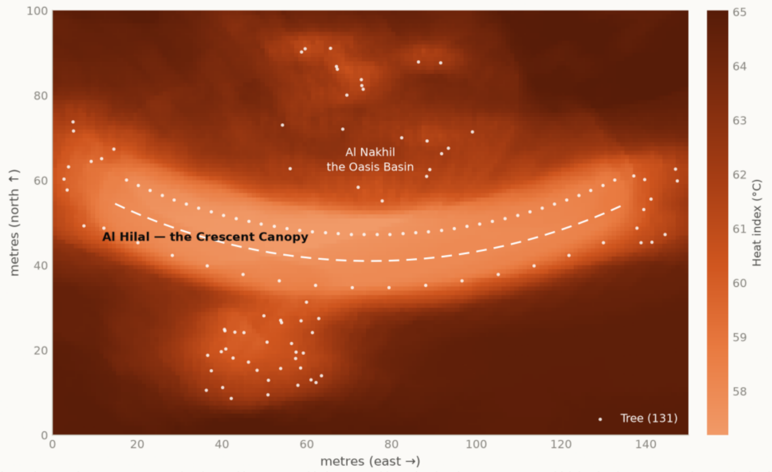
AL SAFA 2 PARK — FALAJ AL SAFA

Dubai Municipality AI Park Design Challenge | Board 2 of 2 — EVIDENCE

What the models measured, and what they changed about the design

Predicted summer comfort across the site

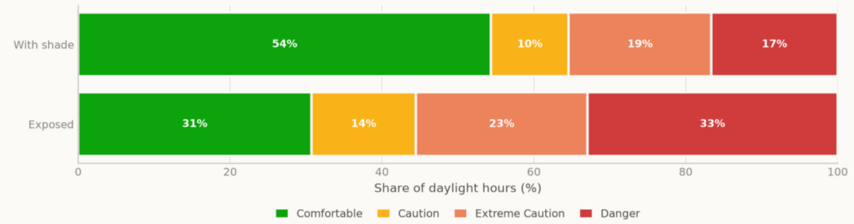
July afternoon heat index per square metre — the shade strategy, tested



Lighter is cooler. The crescent reads as the coolest continuous route across the site, and its concave side is measurably cooler than its convex one — which is why the rooms people are asked to linger in are all placed there. White dots are the 131 trees.
Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; NCM Dubai climate normals 1977-2015 (39-year record) · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Shade converts unusable daylight hours into usable ones

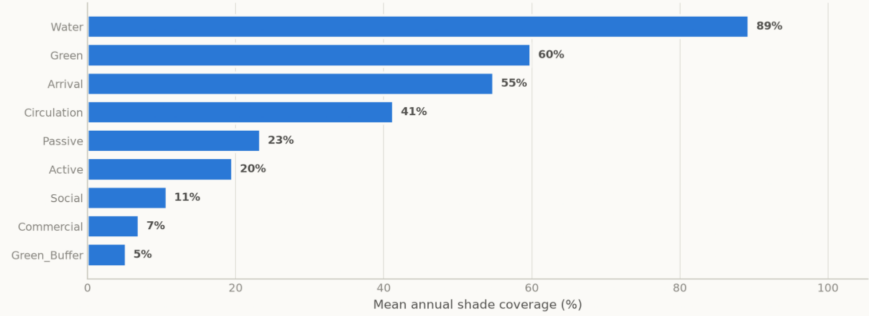
Share of the year's 4,402 daylight hours in each comfort band



Source: NCM Dubai climate normals 1977-2015 (39-year record); comfort bands DERIVED via NWS heat index · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Annual shade coverage by zone type

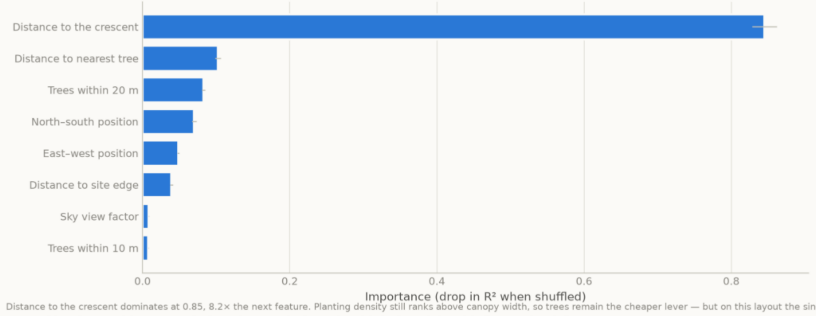
Ray-traced ground-plane occlusion, 1 m grid, sampled across the year



Source: NREL Solar Position Algorithm via pvlib, 25.190N 55.238E; canopy and tree geometry from the Phase 5/6 design · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

What actually determines whether a square metre is shaded

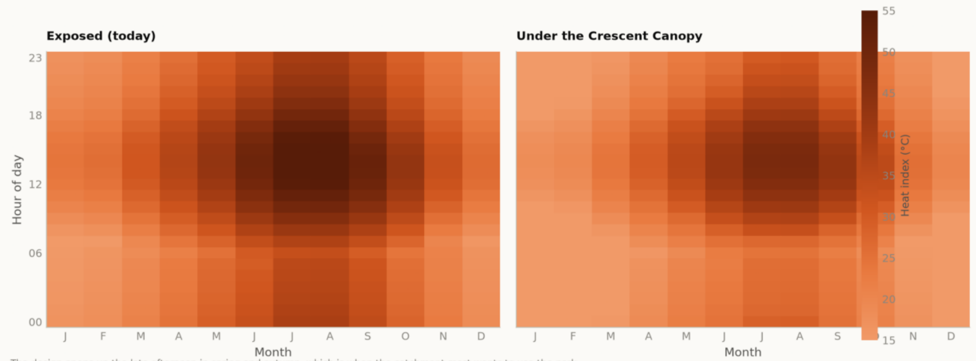
Permutation importance — the drop in accuracy when each feature is shuffled



Distance to the crescent dominates at 0.85, 8.2× the next feature. Planting density still ranks above canopy width, so trees remain the cheaper lever — but on this layout the single strongest predictor of whether a square metre is shaded is simply how far it is from the crescent.
Source: This project's models — see notebooks/ and src/ · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

When is the park comfortable?

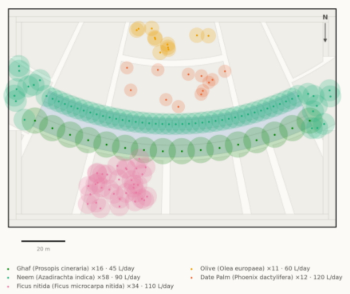
Mean heat index by hour and month — exposed today, shaded as designed



The design opens up the late afternoon in spring and autumn, which is when the catchment most wants to use the park.
Source: NCM Dubai climate normals 1977-2015 (39-year record) — hourly series MODELLED, see src/climate.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Planting plan — 131 trees

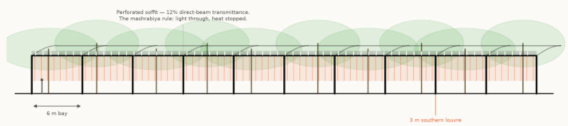
Drawn at mature canopy radius. The avenue is structural: it carries the shoulder seasons the gridshell cannot



28 of 131 trees are UAE natives (Ghaf and Date Palm). Peak summer irrigation demand 11.8 m³/day. Ghaf takes the southern rank because it is the most drought-tolerant species in the schedule and that rank is the more exposed of the two.
Source: data/raw/species_water_carbon_rates.csv; layout from src/solar.py · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Elevation — the Crescent Canopy

60 m of the 124 m run, to scale. 6 m structural bay



21 bays over the full run. Because the plan is an arc, no two bays are identical in plan — but every one is the same section, which is what keeps it buildable. Trees at mature canopy from the Phase 6 planting schedule.
Source: src/config.py CRESCENT; bay spacing from the Phase 6 structure · Al Safa 2 Park · Falaj Al Safa · Dubai Municipality AI Design Challenge

Comfortable daylight hours rise from 44.5% to 64.6% — 20.1 points of the year handed back. Peak heat index falls 56.8 °C → 48.7 °C.
Site-wide mean shade is 34.1%, and that is stated as a position rather than hidden: this scheme makes a few places genuinely excellent instead of the whole site marginally better.
Shade surrogate R^2 0.994 · comfort classifier 97.5% with temperature and humidity withheld · 2 microclimate regimes, k selected by silhouette